

Independent actuator, bias and link
endpoint measurements

Combined plant and controller boundary

Plant

$$Li^2/2 + mv^2/2 + kx^2/2$$

$$v_s = F_s = 0$$

$$P_r = -F_a v > 0$$

actuator reaction

Controller

actuator $L_a i_a^2/2$

bias $L_b i_b^2/2$ and

link $C_d V_d^2/2$

charger off

Winding and mechanical heat
 Ri^2, cv^2

Actuator, bias, converter and logic heat
 $R_a i_a^2, R_b i_b^2, (1 - \eta)P_e, P_0$